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Course Name: SIGNALS AND SYSTEM

Course Code: ECA 0304

Experiment 20 STFT and Wavelet-Based Time-Frequency Analysis of Non-Stationary Signals

OBJECTIVES

• To generate a non-stationary signal containing time-varying frequency components and a transient.

• To compute and visualize the STFT spectrogram.

• To perform wavelet-based time-frequency analysis and interpret the scalogram.

• To compare STFT and wavelet representations for identifying transient and changing frequency components.

Experimental Procedure

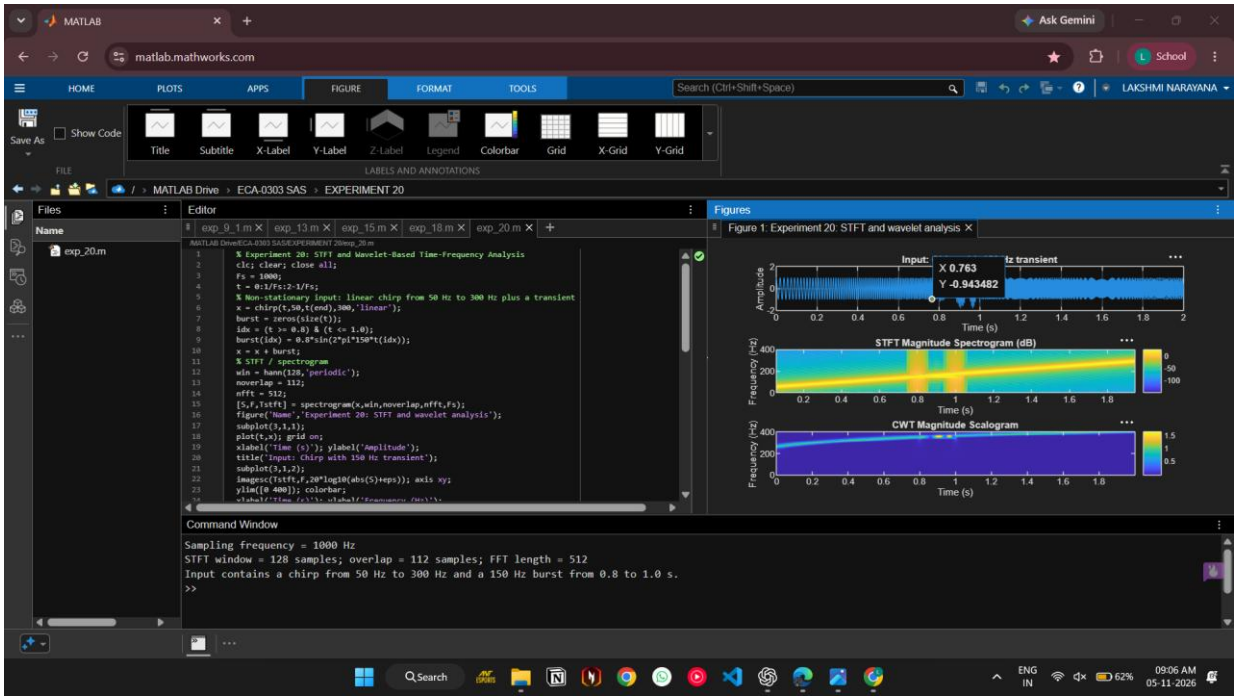
1. Generate a linear chirp from 50 Hz to 300 Hz over 2 seconds.
2. Add a 150 Hz sinusoidal burst between 0.8 s and 1.0 s.
3. Compute the STFT using a 128-sample window, 112-sample overlap, and 512-point FFT.
4. Plot the input signal and STFT magnitude spectrogram.
5. Compute the continuous wavelet transform and plot the scalogram.
6. Compare how both representations show the chirp and transient.

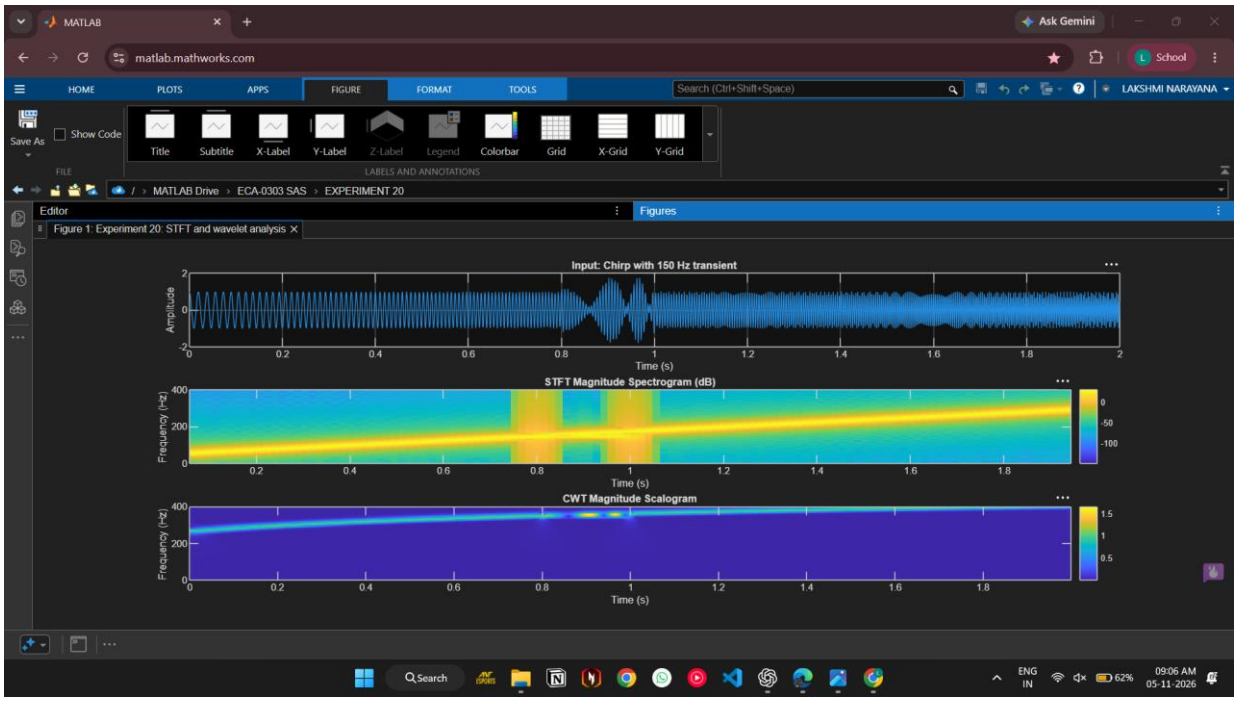
MATLAB Coding

```
% Experiment 20: STFT and Wavelet-Based Time-Frequency Analysis
clc; clear; close all;
Fs = 1000;
t = 0:1/Fs:2-1/Fs;
% Non-stationary input: linear chirp from 50 Hz to 300 Hz plus a transient
x = chirp(t,50,t(end),300,'linear');
burst = zeros(size(t));
idx = (t >= 0.8) & (t <= 1.0);
burst(idx) = 0.8*sin(2*pi*150*t(idx));
x = x + burst;
```

```
% STFT / spectrogram
win = hann(128,'periodic');
noverlap = 112;
nfft = 512;
[S,F,Tstft] = spectrogram(x,win,noverlap,nfft,Fs);
figure('Name','Experiment 20: STFT and wavelet analysis');
subplot(3,1,1);
plot(t,x); grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Input: Chirp with 150 Hz transient');
subplot(3,1,2);
imagesc(Tstft,F,20*log10(abs(S)+eps)); axis xy;
ylim([0 400]); colorbar;
xlabel('Time (s)'); ylabel('Frequency (Hz)');
title('STFT Magnitude Spectrogram (dB)');
% Continuous wavelet transform; requires Wavelet Toolbox
subplot(3,1,3);
[wt,fCWT] = cwt(x,Fs);
imagesc(t,fCWT,abs(wt)); axis xy;
ylim([0 400]); colorbar;
xlabel('Time (s)'); ylabel('Frequency (Hz)');
title('CWT Magnitude Scalogram');
% Report settings used for reproducibility
fprintf('Sampling frequency = %d Hz\n',Fs);
fprintf('STFT window = %d samples; overlap = %d samples; FFT length = %d\n', ...
length(win),noverlap,nfft);
fprintf('Input contains a chirp from 50 Hz to 300 Hz and a 150 Hz burst from 0.8 to 1.0 s.\n');
```

Input and Output Graphs


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Files
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exp_20.m
Editor
exp_9_1.m
exp_13.m
exp_15.m
exp_18.m
exp_20.m
% Experiment 20: STFT and Wavelet-Based Time-Frequency Analysis
clc; clear; close all;
fs = 1000;
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burst = zeros(size(t));
idx = (t >= 0.8) & (t <= 1.0);
burst(idx) = 0.8*sin(2*pi*150*t(idx));
x = x + burst;
% STFT / spectrogram
win = hann(128,'periodic');
noverlap = 112;
nfft = 512;
[S,f,Tstft] = spectrogram(x,win,noverlap,nfft,fs);
figure('Name','Experiment 20: STFT and wavelet analysis');
subplot(3,1,1);
plot(t,x); grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Input: Chirp with 150 Hz transient');
subplot(3,1,2);
imagesc(Tstft,f,20*log10(abs(S)+eps)); axis xy;
ylim([0 400]); colorbar;
Figures
Figure 1: Experiment 20: STFT and wavelet analysis
Input:
Hz transient
X 0.763
Y -0.943482
Amplitude
Time (s)
STFT Magnitude Spectrogram (dB)
Frequency (Hz)
CWT Magnitude Scalogram
Command Window
Sampling frequency = 1000 Hz
STFT window = 128 samples; overlap = 112 samples; FFT length = 512
Input contains a chirp from 50 Hz to 300 Hz and a 150 Hz burst from 0.8 to 1.0 s.
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